**Prescribing Information :** For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only.

# GABAPIN-300

(Gabapentin Capsules 300 mg)

**COMPOSITION**

Each hard gelatin capsule contains:
Gabapentin 300 mg

**DESCRIPTION**

Gabapentin is described as 1-(aminomethyl) cyclohexanacetic acid with an empirical formula of $C_9H_{17}NO_2$ and a molecular weight of 171.24.

Gabapentin 300 mg capsule is Opaque yellow / opaque yellow colored size "1" hard gelatin capsules, imprinted with "G300" on cap with blue ink, containing white to off white granular powder.

**CLINICAL PHARMACOLOGY**

**Mechanism of Action**

The mechanism by which gabapentin exerts its anticonvulsant action is unknown, but in animal test systems designed to detect anticonvulsant activity, gabapentin prevents seizures as do other marketed anticonvulsants. Gabapentin exhibits antiseizure activity in mice and rats in both the maximal electroshock and pentylenetetrazole seizure models and other preclinical models (*e.g.,* strains with genetic epilepsy, etc.). The relevance of these models to human epilepsy is not known.
Gabapentin is structurally related to the neurotransmitter GABA (gamma-aminobutyric acid) but it does not interact with GABA receptors, it is not converted metabolically into GABA or a GABA agonist, and it is not an inhibitor of GABA uptake or degradation. Gabapentin was tested in radioligand binding assays at concentrations up to 100 mcM and did not exhibit affinity for a number of other common receptor sites, including benzodiazepine, glutamate, N-methyl-D-aspartate (NMDA), quisqualate, kainate, strychnine-insensitive or strychnine-sensitive glycine, alpha 1, alpha 2, or beta adrenergic, adenosine A1 or A2, cholinergic muscarinic or nicotinic, dopamine D1 or D2, histamine H1, serotonin S1 or S2, opiate mu, delta or kappa, voltage-sensitive calcium channel sites labeled with nitrendipine or diltiazem, or at voltage-sensitive sodium channel sites with batrachotoxinin A 20-alpha-benzoate.
Several test systems ordinarily used to assess activity at the NMDA receptor have been examined. Results are contradictory. Accordingly, no general statement about the effects, if any, of gabapentin at the NMDA receptor can be made.
*In vitro* studies with radiolabeled gabapentin have revealed a gabapentin binding site in areas of rat brain including neocortex and hippocampus. The identity and function of this binding site remain to be elucidated.

**Pharmacokinetics and Drug Metabolism**

All pharmacological actions following gabapentin administration are due to the activity of the parent compound; gabapentin is not appreciably metabolized in humans.
***Oral Bioavailability*:** Gabapentin bioavailability is not dose proportional; *i.e.,* as dose is increased, bioavailability decreases. A 400-mg dose, for example is about 25% less bioavailable than a 100-mg dose. Over the recommended dose range of 300 to 600 mg t.i.d., however, the differences in bioavailability are not large, and bioavailability is about 60 percent. Food has no effect on the rate and extent of absorption of gabapentin.
***Distribution*:** Gabapentin circulates largely unbound (<3%) to plasma protein. The apparent volume of distribution of gabapentin after 150 mg intravenous administration is 58±6 L (Mean ±SD). In patients with epilepsy, steady-state predose ($C_{min}$) concentrations of gabapentin in cerebrospinal fluid were approximately 20% of the corresponding plasma concentrations.
***Elimination*:** Gabapentin is eliminated from the systemic circulation by renal excretion as unchanged drug. Gabapentin is not appreciably metabolized in humans.
Gabapentin elimination half-life is 5 to 7 hours and is unaltered by dose or following multiple dosing. Gabapentin elimination rate constant, plasma clearance, and renal clearance are directly proportional to creatinine clearance (see Special Populations, Patients with Renal Insufficiency). In elderly patients, and in patients with impaired renal function, gabapentin plasma clearance is reduced. Gabapentin can be removed from plasma by hemodialysis.

**INDICATIONS**

**Epilepsy:**

Gabapentin is indicated as adjunctive therapy in the treatment of partial seizures with and without secondary generalization in adults and children age 3 years and above. Safety and effectiveness for adjunctive therapy in pediatric patients below the age of 3 years have not been established.

**Neuropathic pain:**

Gabapentin is indicated for the treatment of neuropathic pain which includes diabetic neuropathy, post-herpetic neuralgia and trigeminal neuralgia in adults age 18 years and above. Safety and effectiveness in patients below the age of 18 years have not been established.

**DOSAGE AND ADMINISTRATION**

**General**

Gabapentin is given orally with or without food. When in the judgement of the clinician there is a need for dose reduction, discontinuation, or substitution with an alternative medication, this should be done gradually over a minimum of one week.

**Epilepsy**

Adults and pediatric patients over 12 years of age:
The effective dosing range was 900 to 3600 mg/day. Therapy may be initiated by administering 300mg three times a day (TID) on Day 1 or by titrating the dose as described in TABLE 1. Thereafter, the dose can be increased in three equally divided doses up to a maximum dose of 3600 mg/day. Dosages up to 4800 mg/day have been well tolerated. The maximum time between doses in the three times a day TID schedule should not exceed 12 hours to prevent breakthrough convulsions.

| Table 1 | | | |
|---|---|---|---|
| Dosing Chart – Initial Titration | | | |
| Dose | Day 1 | Day 2 | Day 3 |
| 900 mg | 300 mg once daily | 300 mg two times daily | 300 mg three times daily |

Pediatric patients age 3-12 years:
The starting dose should range from 10 to 15 mg/kg/day given in equally divided doses (three times a day), and the effective dose reached by upward titration over a period of approximately three days. The effective dose of gabapentin in pediatric patients age 5 years and older is 25 to 35 mg/kg/day given in equally divided doses (three times a day). The effective dose in pediatric patients ages 3 to less than 5 years is 40 mg/kg/day given in equally divided doses (three times a day). Dosages up to 50 mg/kg/day have been well tolerated. The maximum time interval between doses should not exceed 12 hours.

It is not necessary to monitor gabapentin plasma concentrations to optimize gabapentin therapy. Further, gabapentin may be used in combination with other antiepileptic drugs without concern for alteration of the plasma concentrations of gabapentin or serum concentrations of other antiepileptic drugs.

**Neuropathic pain in adults:**

The starting dose is 900 mg/day given as three equally divided doses, and increased if necessary, based on response, up to a maximum dose of 3600 mg/day. Therapy should be initiated by titrating the dose as described in Table 1.

**Dosage adjustment in Impaired Renal Functions Patients with neuropathic pain or epilepsy:**

Dosage adjustment is recommended in patients with compromised renal function as described in TABLE 2 and/or those undergoing hemodialysis.

| Table 2 | |
|---|---|
| Dosage of Gabapentin in Adults Based on Renal Function | |
| Creatinine Clearance (ml/min) | Total Daily Dose (mg/day) |
| 80 or more | 900 – 3600 |
| 50 - 79 | 600 – 1800 |
| 30 - 49 | 300 - 900 |
| 15 - 29 | 150 (to be administered as 300 mg every other day)-600 |
| < 15 | 150 (to be administered as 300 mg every other day)-300 |

* Total daily dose should be administered as a TID regimen. Doses used to treat patients with normal renal function (creatinine clearance >80 ml/min) range from 900 to 3600 mg/day. Reduced dosages are for patients with renal impairment (creatinine clearance < 79 ml/min).

**Dosage adjustment in patients undergoing hemodialysis**

For patients undergoing hemodialysis who have never received gabapentin, a loading dose of 300 to 400 mg is recommended, then 300 mg of gabapentin following each 4 hours of hemodialysis.

**SIDE EFFECTS**

The most commonly observed adverse events associated with the use of gabapentin in adults, not seen at an equivalent frequency among placebo-treated patients, were dizziness, somnolence, and peripheral edema.
In controlled studies in post-herpetic neuralgia, 16% of patients who received gabapentin and 9% of the patients who received placebo discontinued treatment because of an adverse event. The adverse events that most frequently led to withdrawal were dizziness, somnolence, and nausea.
The most commonly observed adverse events associated with the use of gabapentin in combination with other antiepileptic drugs in patients >12 years of age were somnolence, dizziness, ataxia, fatigue and nystagmus.

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Approximately 7% of the individuals who received gabapentin in discontinued treatment because of an adverse event. The adverse events most commonly associated with withdrawal were somnolence (1.2%), ataxia (0.8%), fatigue (0.6%), nausea and/or vomiting (0.6%), and dizziness (0.6%).

***Body as a Whole:*** *Frequent*: Asthenia, malaise, face edema; *Infrequent*: Allergy, generalized edema, weight decrease, chill; *Rare*: Strange feelings, lassitude, alcohol intolerance, hangover effect.
***Cardiovascular System:*** *Frequent*: Hypertension; *Infrequent*: Hypotension, angina pectoris, peripheral vascular disorder, palpitation, tachycardia, migraine, murmur; *Rare*: Atrial fibrillation, heart failure, thrombophlebitis, deep thrombophlebitis, myocardial infarction, cerebrovascular accident, pulmonary thrombosis, ventricular extrasystoles, bradycardia, premature atrial contraction, pericardial rub, heart block, pulmonary embolus, hyperlipidemia, hypercholesterolemia, pericardial effusion, pericarditis.
***Digestive System:*** *Frequent*: Anorexia, flatulence, gingivitis; *Infrequent*: Glossitis, gum hemorrhage, thirst, stomatitis, increased salivation, gastroenteritis, hemorrhoids, bloody stools, fecal incontinence, hepatomegaly; *Rare*: Dysphagia, eructation, pancreatitis, peptic ulcer, colitis, blisters in mouth, tooth discolor, perleche, salivary gland enlarged, lip hemorrhage, esophagitis, hiatal hernia, hematemesis, proctitis, irritable bowel syndrome, rectal hemorrhage, esophageal spasm.
***Endocrine System:*** *Rare*: Hyperthyroid, hypothyroid, goiter, hypoestrogen, ovarian failure, epididymitis, swollen testicle, cushingoid appearance.
***Hematologic and Lymphatic System:*** *Frequent*: Purpura most often described as bruises resulting from physical trauma; *Infrequent*: Anemia, thrombocytopenia, lymphadenopathy; *Rare*: WBC count increased, lymphocytosis, non-Hodgkin's lymphoma, bleeding time increased.
***Musculoskeletal System:*** *Frequent*: Arthralgia; *Infrequent*: Tendinitis, arthritis, joint stiffness, joint swelling, positive Romberg test; *Rare*: Costochondritis, osteoporosis, bursitis, contracture.
***Nervous System:*** *Frequent*: Vertigo, hyperkinesia, paresthesia, decreased or absent reflexes, increased reflexes, anxiety, hostility; *Infrequent*: CNS tumors, syncope, dreaming abnormal, aphasia, hypesthesia, intracranial hemorrhage, hypotonia, dysesthesia, paresis, dystonia, hemiplegia, facial paralysis, stupor, cerebellar dysfunction, positive Babinski sign, decreased position sense, subdural hematoma, apathy, hallucination, decrease or loss of libido, agitation, paranoia, depersonalization, euphoria, feeling high, doped-up sensation, suicidal, psychosis; *Rare*: Choreoathetosis, orofacial dyskinesia, encephalopathy, nerve palsy, personality disorder, increased libido, subdued temperament, apraxia, fine motor control disorder, meningismus, local myoclonus, hyperesthesia, hypokinesia, mania, neurosis, hysteria, antisocial reaction, suicide gesture.
***Respiratory System:*** *Frequent*: Pneumonia; *Infrequent*: Epistaxis, dyspnea, apnea; *Rare*: Mucositis, aspiration pneumonia, hyperventilation, hiccup, laryngitis, nasal obstruction, snoring, bronchospasm, hypoventilation, lung edema, respiratory depression.
***Dermatological:*** *Infrequent*: Alopecia, eczema, dry skin, increased sweating, urticaria, hirsutism, seborrhea, cyst, herpes simplex; *Rare*: Herpes zoster, skin discolor, skin papules, photosensitive reaction, leg ulcer, scalp seborrhea, psoriasis, desquamation, maceration, skin nodules, subcutaneous nodule, melanosis, skin necrosis, local swelling.
***Urogenital System:*** *Infrequent*: Hematuria, dysuria, urination frequency, cystitis, urinary retention, urinary incontinence, vaginal hemorrhage, amenorrhea, dysmenorrhea, menorrhagia, breast cancer, unable to climax, ejaculation abnormal; *Rare*: Kidney pain, leukorrhea, pruritus genital, renal stone, acute renal failure, anuria, glycosuria, nephrosis, nocturia, pyuria, urination urgency, vaginal pain, breast pain, testicle pain.
***Special Senses:*** *Frequent*: Abnormal vision; *Infrequent*: Cataract, conjunctivitis, eyes dry, eye pain, visual field defect, photophobia, bilateral or unilateral ptosis, eye hemorrhage, hordeolum, hearing loss, earache, tinnitus, inner ear infection, otitis, taste loss, unusual taste, eye twitching, ear fullness; *Rare*: Eye itching, abnormal accommodation, perforated ear drum, sensitivity to noise, eye focusing problem, watery eyes, retinopathy, glaucoma, iritis, corneal disorders, lacrimal dysfunction, degenerative eye changes, blindness, retinal degeneration, miosis, chorioretinitis, strabismus, eustachian tube dysfunction, labyrinthitis, otitis externa, odd smell.

**POSTMARKETING EXPERIENCE**
Dysphagia

**DRUG ABUSE AND DEPENDENCE**
The abuse and dependence potential of gabapentin has not been evaluated in human studies.

**DRUG INTERACTIONS**
Patients should be carefully observed for signs of CNS depression, such as somnolence, and the dose of gabapentin or morphine should be reduced appropriately when gabapentin is co-administered with morphine. No interaction between gabapentin and Phenobarbital, phenytoin, valproic acid, or carbamazepine has been observed.
Co-administration of Gabapentin with oral contraceptives containing norethindrone and/or ethinyl estradiol does not influence the steady-state pharmacokinetics of either component.
Co-administration of gabapentin with antacids containing aluminium and magnesium, reduces gabapentin bioavailability up to 24%. It is recommended that gabapentin be taken at the earliest two hours following antacid administration.
Renal excretion of gabapentin is unaltered by probenecid.
A slight decrease in renal excretion of gabapentin that is observed when it is co-administered with cimetidine is not expected to be clinical importance.

**CONTRAINDICATIONS**
Gabapentin is contraindicated in patients with known hypersensitivity to it.

**WARNINGS & PRECAUTIONS**
If a patient develops acute pancreatitis under treatment with gabapentin, discontinuation of gabapentin should be considered. Although there is no evidence of rebound seizures with gabapentin, abrupt withdrawal of anticonvulsant agents in epileptic patients may precipitate status epilepticus. As with other antiepileptic medicinal products, some patients may experience an increase in seizure frequency or the onset of new types of seizures with gabapentin. As with other anti-epileptics, attempts to withdraw concomitant anti-epileptics in treatment refractive patients on more than one anti-epileptics, in order to reach gabapentin monotherapy have a low success rate.
Gabapentin is not considered effective against primary generalized seizures such as absences and may aggravate these seizures in some patients. Therefore, gabapentin should be used with caution in patients with mixed seizures including absences. No systematic studies in patients 65 years or older have been conducted with gabapentin. Somnolence, peripheral oedema and asthenia occurred in a somewhat higher percentage in patients aged 65 years or above, than in younger patients. The effects of long-term (greater than 36 weeks) gabapentin therapy on learning, intelligence, and development in children and adolescents have not been adequately studied. The benefits of prolonged therapy must therefore be weighed against the potential risks of such therapy.

Antiepileptic drugs (AEDs), including Gabapentin, has potential for an increase in risk of suicidal thoughts or behaviors in patients taking these drugs for any indication.

Respiratory depression
Gabapentin has been associated with severe respiratory depression. Patients with compromised respiratory function, respiratory or neurological disease, renal impairment, concomitant use of central nervous system (CNS) depressants and the elderly might be at higher risk of experiencing this severe adverse reaction. Dose adjustments might be necessary in these patients.

**PREGNANCY & LACTATION**
**Pregnancy**
There are no adequate and well-controlled studies in pregnant women. This drug should be used during pregnancy only if the potential benefit to the patient justifies the potential risk to the fetus.

**Lactation**
Gabapentin is excreted in human milk. Because the effect on the nursing infant is unknown, caution should be exercised when gabapentin is administered to a nursing mother. Gabapentin should be used in nursing mothers only if the benefits clearly outweigh the risks.

**OVERDOSE**
A lethal dose of gabapentin was not identified in mice and rats receiving single oral doses as high as 8000 mg/kg. Signs of acute toxicity in animals included ataxia, labored breathing, ptosis, sedation, hypoactivity, or excitation.
Acute oral overdoses of gabapentin up to 49 grams have been reported. In these cases, double vision, slurred speech, drowsiness, lethargy and diarrhea were observed. All patients recovered with supportive care.
Gabapentin can be removed by hemodialysis. Although hemodialysis has not been performed in the few overdose cases reported, it may be indicated by the patient's clinical state or in patients with significant renal impairment.

**STORAGE**
Store below 30°C.

**PRESENTATION**
GABAPIN-300 capsules are available in alu-alu blister of 10 capsules. 5 such blisters are packed in a printed box. (5 x 10/box)


Manufactured by:
INTAS
***INTAS PHARMACEUTICALS LTD.***
Plot No. 457 & 458, Village Matoda, Bavla road, And Plot No. 191/218 P, Village Chacharwadi,Ta -Sanand, Dist. Ahmedabad - 382210, India

**Importer**
Jetpharma Sdn Bhd - Selangor, MALAYSIA



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